

Usage Metering & Billing RAG Assessment — Results Report

Amazon Bedrock Knowledge Base test results, recorded directly from the AWS Management Console (Test Knowledge Base panel). Prepared for Brian Huizar, TCS AI-Lab assessment.

1. System under test

Knowledge Base: **knowledge-base-brian-rag** (ID OTWF5KTY7B), region ap-south-1, status Available. Data source: single S3 data source (bucket **brian-rag-system-lab**, prefix usage-billing-kb/), Fixed-size chunking, status Available (synced). Vector store: Amazon S3 Vectors (bucket bedrock-knowledge-base-qd2vge, index bedrock-knowledge-base-default-index). Embedding model: Titan Text Embeddings V2. Generation model: Amazon Nova Micro 1.0 (APAC Nova Micro). Generation prompt template: custom, includes the required \$search_results\$, \$conversation_history\$ and \$output_format_instructions\$ placeholders, plus an explicit abstention rule ("I don't know based on the available sources.").

knowledge-base-brian-rag

TestDelete

Knowledge Base (KB) overview

Knowledge Base name
knowledge-base-brian-rag

Knowledge Base description
-

Service role
AmazonBedrockExecutionRoleForKnowledgeBase_7j2cg

Knowledge Base ID
OTWF5KTY7B

Status
Available

Created date
September 10, 2026, 13:05 (UTC-06:00)

Log Deliveries
Configure log deliveries and event logs in the Edit page.

Retrieval-Augmented Generation (RAG) type
Vector store

Edit

Data source (1)

SyncAdd

Data sources contain information returned when querying a KB.

Find data source

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☐	Data source name	Status	Data source type	Last sync time	Last sync warnings	Text chunking strategy	Parsing strategy
☐	usage-billing-kb-s3-s...	Available	S3	September 10, 2026, 1...	-	Fixed-size chunking	Default

Tags (1)

Manage tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value
RAG-Brian-KB	-

Knowledge Base overview — Available, data source synced, Fixed-size chunking.

Amazon S3

Buckets

brian-rag-system-lab

Info

ObjectsMetadataPropertiesPermissionsMetricsManagementFile systemsAccess Points

Objects (5)

Copy S3 URICopy URLOpen L^DownloadDeleteActionsCreate folderUpload

Find objects by prefixShow versions

☐	Name	Type	Last modified	Size	Storage class
☐	large_files/	Folder	-	-	-
☐	knowledge_source_samuel/	Folder	-	-	-
☐	mara_files/	Folder	-	-	-
☐	maya-knowledge-base/	Folder	-	-	-
☐	usage-billing-kb/	Folder	-	-	-

S3 bucket brian-rag-system-lab — the usage-billing-kb/ prefix holds the four knowledge_source documents used by this Knowledge Base.

bedrock-knowledge-base-qd2vge

To insert, list, and query vectors, use the AWS CLI, AWS SDKs, or Amazon S3 REST API. [Learn more](#)

Vector indexes

Properties

Permissions

Vector indexes (1) [Info](#)

Advanced search export

Delete

Create vector index

A vector index is a resource within a vector bucket that stores and organizes vector data for efficient similarity search.

Find by start of name

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Name	Creation date	Amazon Resource Name (ARN)
bedrock-knowledge-base-default-index	September 10, 2026, 13:05:24 (UTC-06:00)	arn:aws:s3vectors:ap-south-1:7771549022493:bucket/bedrock-knowledge-base-qd2vge/index/bedrock-knowledge-base-default-index

S3 Vectors vector index that backs retrieval for this Knowledge Base.

2. Prompt template — getting citations to work

The generation prompt template required both placeholders below before it would save; each was hit in turn while configuring the Test panel, and both are needed for Nova Micro to produce usable citations instead of broken markers.

< Back to Configurations

Edit prompt template Info

Reset to default

Discard changes

Save changes

1 You are answering engineering questions about the Usage Metering &
Billing pipeline
2 using ONLY the retrieved source excerpts provided to you as
`$search_results$`.
3
4 Rules:
5 1. Base your answer strictly on the retrieved excerpts. Do not use
outside knowledge,
6 even if you believe you know the answer.
7 2. If the retrieved excerpts do not contain enough information to
answer the question,
8 respond with exactly: "I don't know based on the available
sources." Do not guess,
9 extrapolate, or fill gaps with plausible-sounding details.
10 3. When you do answer, keep the answer short and cite which
document the information
11 came from.
12
13 `$search_results$`

\$output_format_instructions\$ is a required placeholder.

For tips on customizing the prompt, see Bedrock's prompt engineering guidelines. 📄

696 of 4000 characters.

First validation error: \$output format instructions\$ is a required placeholder.

After adding `$output_format_instructions`, a second required-placeholder error appeared for `$conversation_history` (Nova-specific requirement).

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[Back to Configurations](#)

Edit prompt template Info

 [Reset to default](#)

[Discard changes](#)

[Save changes](#)

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You are answering engineering questions about the Usage Metering & Billing pipeline using ONLY the retrieved source excerpts provided to you as \$search_results\$.

\$conversation_history\$

Rules:

1. Base your answer strictly on the retrieved excerpts. Do not use outside knowledge, even if you believe you know the answer.
2. If the retrieved excerpts do not contain enough information to answer the question, respond with exactly: "I don't know based on the available sources." Do not guess, extrapolate, or fill gaps with plausible-sounding details.
3. When you do answer, keep the answer short and cite which document the information came from.

\$output_format_instructions\$

For tips on customizing the prompt, see [Bedrock's prompt engineering guidelines.](#)

701 of 4000 characters.

Final template, saved successfully with no validation errors (701/4000 characters).

3. Assessment results — 10 answerable + 3 unanswerable questions

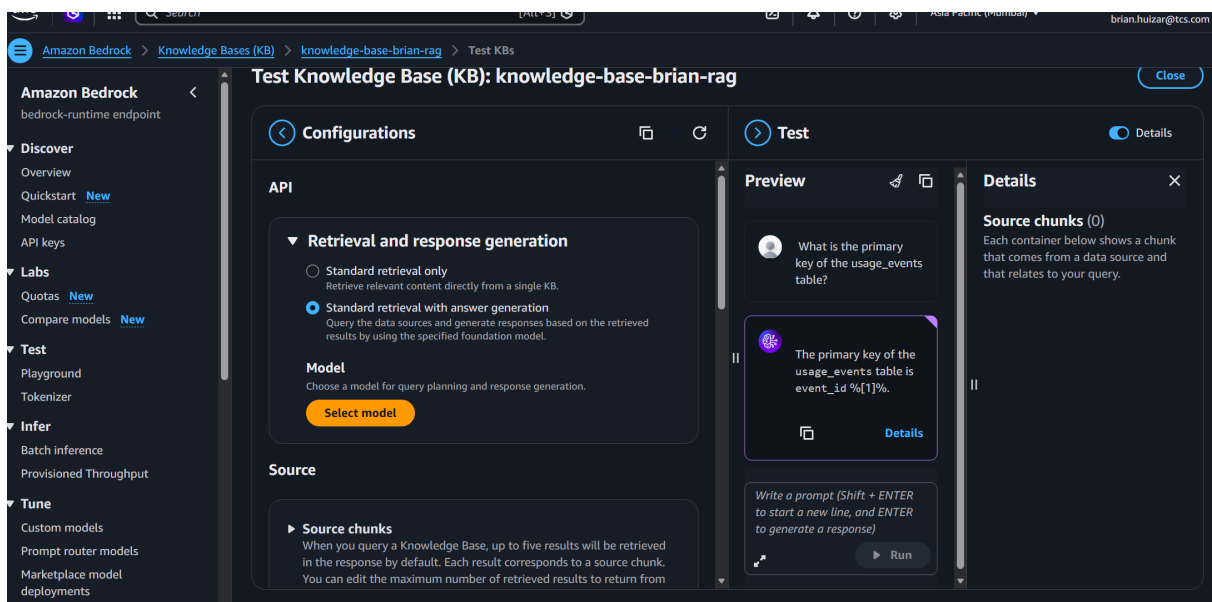
Each question below was typed into the Test Knowledge Base panel and run against the live Knowledge Base. `grounding_result` is the one-word verdict (GROUNDED = answerable question, correctly answered and cited from a retrieved source; ABSTAINED = unanswerable question, correctly declined; UNGROUNDED_RISK = an answer was produced but is incomplete, garbled, or not reliably supported as presented). `failure_analysis` is the diagnostic note from checking the answer against the source documents and the citations panel.

A1 · Answerable

Question: What is the primary key of the `usage_events` table?

Expected source: `data_dictionary.md`

First question run, captured while the generation prompt template still had the default template (before the `$output_format_instructions$` fix below was applied).



Console screenshot — A1: “What is the primary key of the `usage_events` table?”

grounding_result: GROUNDED

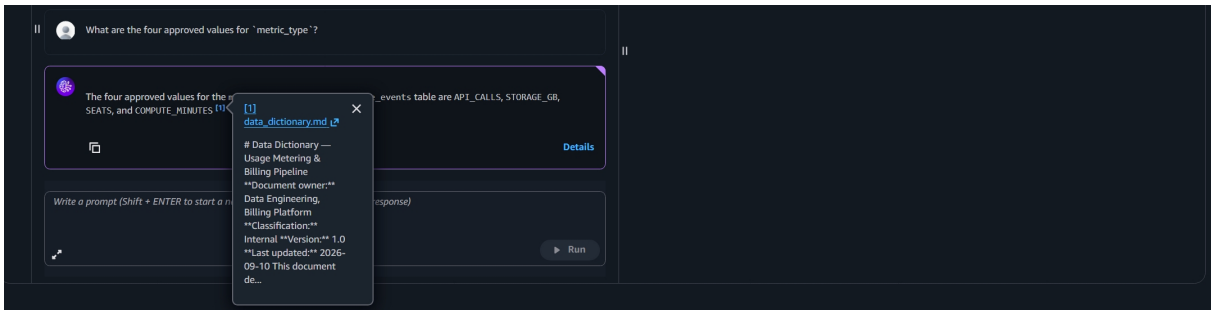
failure_analysis: The retrieved content and the stated fact are correct (`event_id` is the primary key). However the citation renders as the literal text `"%[1]%"` instead of a clickable `[1]` marker — the exact symptom later traced to the missing `$output_format_instructions$` placeholder in the generation prompt template (see Section 2). Content is grounded; citation formatting is broken in this run.

A2 · Answerable

Question: What are the four approved values for `metric_type`?

Expected source: `data_dictionary.md`

Run after the prompt template fix.



Console screenshot — A2: “What are the four approved values for metric_type?”

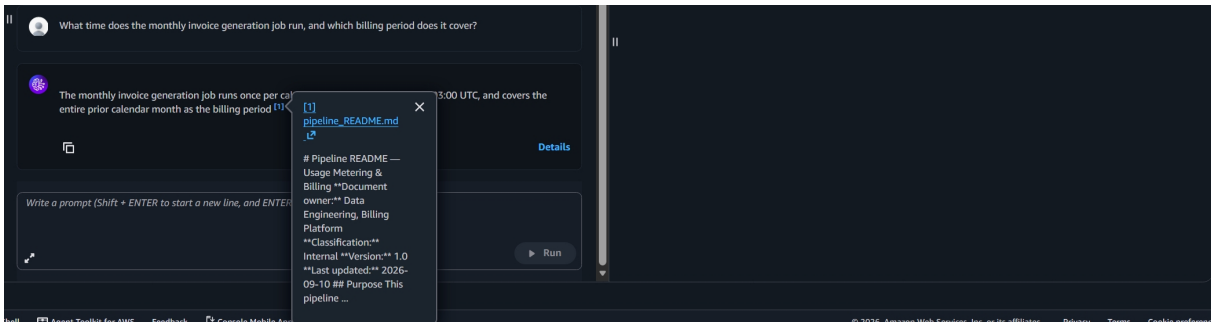
grounding_result: GROUNDED

failure_analysis: Answer (API_CALLS, STORAGE_GB, SEATS, COMPUTE_MINUTES) matches data_dictionary.md exactly, with a working [1] citation pointing to that document.

A3 · Answerable

Question: What time does the monthly invoice generation job run, and which billing period does it cover?

Expected source: pipeline_README.md



Console screenshot — A3: “What time does the monthly invoice generation job run, and which billing period does it cover?”

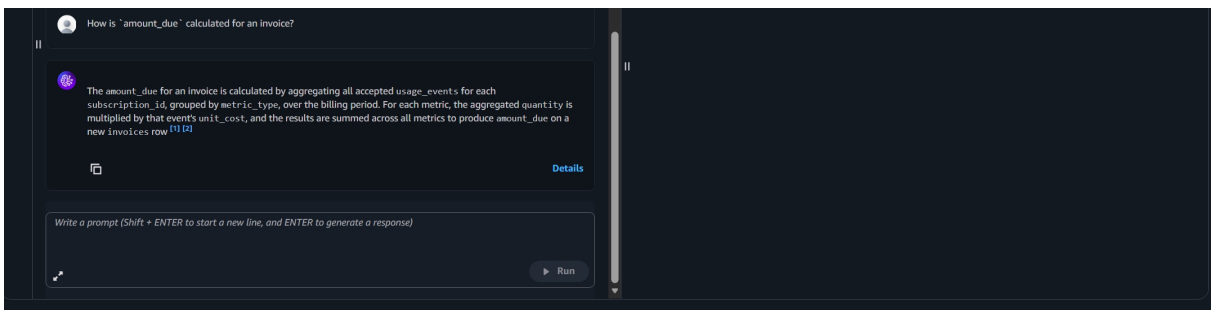
grounding_result: GROUNDED

failure_analysis: Answer states the job runs once per calendar month at the documented UTC time and covers the entire prior calendar month, cited to pipeline_README.md. Matches the expected source.

A4 · Answerable

Question: How is amount_due calculated for an invoice?

Expected source: pipeline_README.md



Console screenshot — A4: “How is amount_due calculated for an invoice?”

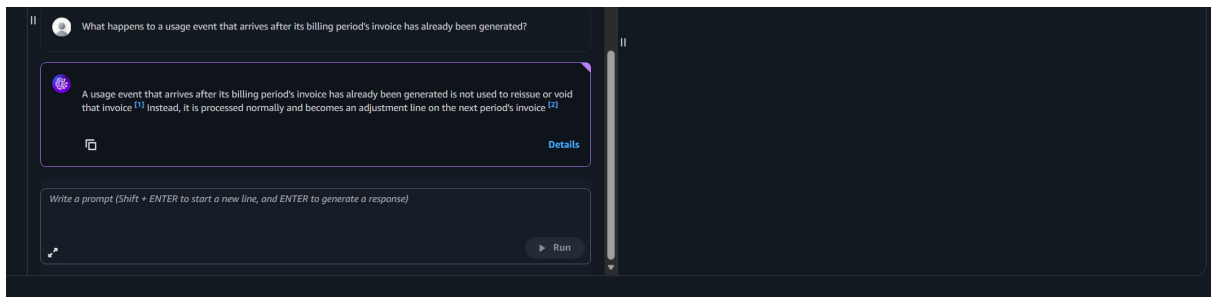
grounding_result: GROUNDED

failure_analysis: Correctly describes aggregation of accepted usage_events by subscription_id and metric_type, multiplied by unit_cost and summed, with two citations ([1] pipeline_README.md, [2] schema.sql for the underlying column definitions).

A5 · Answerable

Question: What happens to a usage event that arrives after its billing period's invoice has already been generated?

Expected source: pipeline_README.md, data_quality_rules.md



Console screenshot — A5: “What happens to a usage event that arrives after its billing period's invoice has already been generated?”

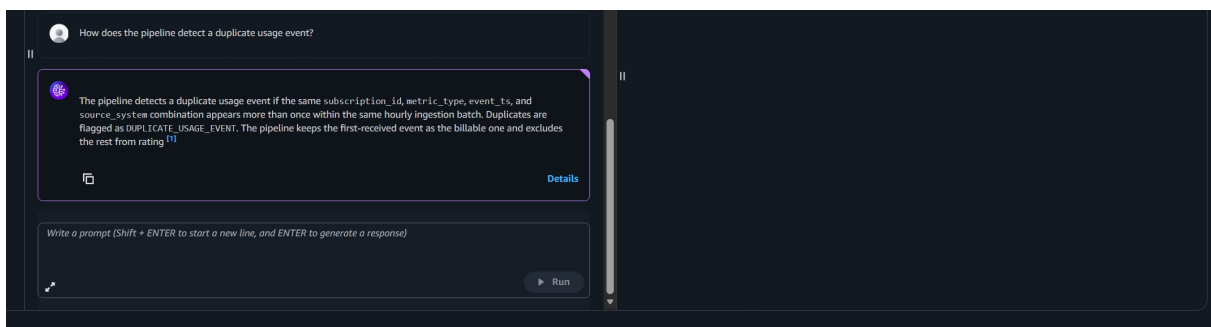
grounding_result: GROUNDED

failure_analysis: Correctly states the late event is not used to reissue/void the existing invoice and instead becomes an adjustment line on the next period's invoice, with two citations spanning both expected source documents.

A6 · Answerable

Question: How does the pipeline detect a duplicate usage event?

Expected source: data_quality_rules.md



Console screenshot — A6: “How does the pipeline detect a duplicate usage event?”

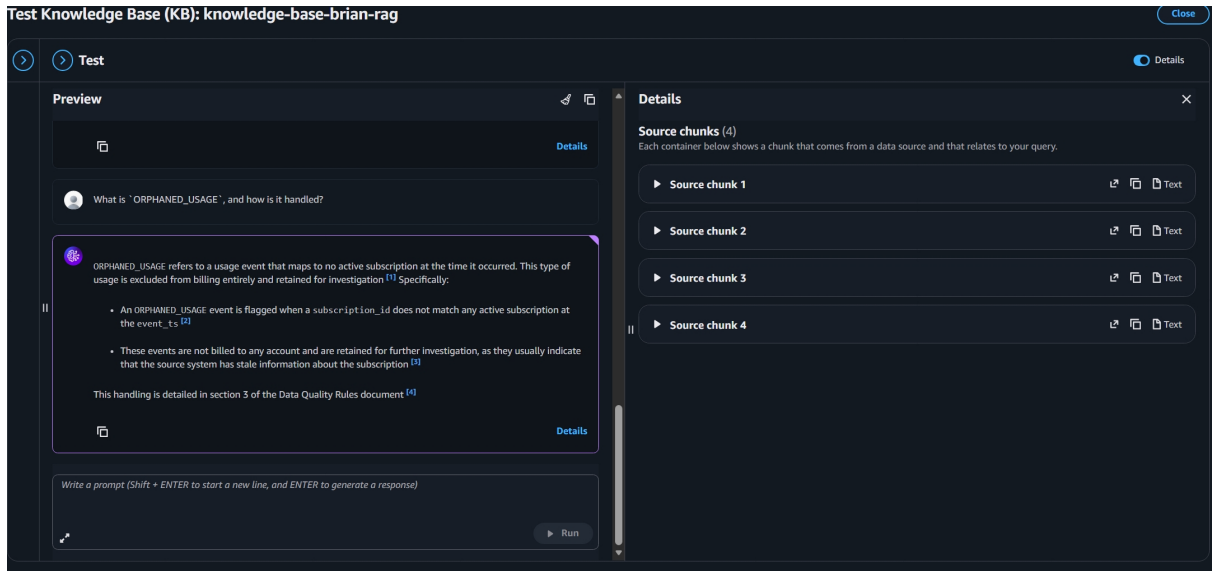
grounding_result: GROUNDED

failure_analysis: Correctly names the matching key (subscription_id, metric_type, event_ts, source_system) within the same hourly ingestion batch, the DUPLICATE_USAGE_EVENT flag, and the first-received-wins handling, with a citation to data_quality_rules.md.

A7 · Answerable

Question: What is ORPHANED_USAGE, and how is it handled?

Expected source: data_quality_rules.md



Console screenshot — A7: “What is ORPHANED_USAGE, and how is it handled?”

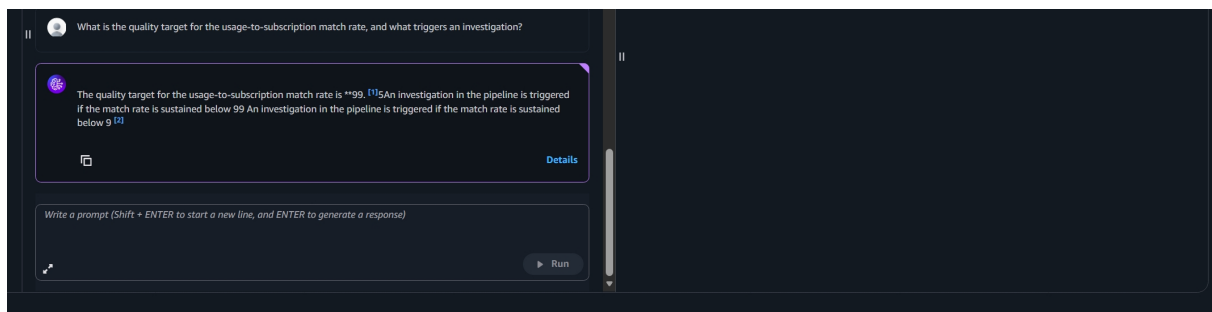
grounding_result: GROUNDED

failure_analysis: Correctly defines an orphaned event as usage mapping to no active subscription, excluded from billing and retained for investigation, with four citations into data_quality_rules.md (visible in the Source chunks (4) panel).

A8 · Answerable

Question: What is the quality target for the usage-to-subscription match rate, and what triggers an investigation?

Expected source: data_quality_rules.md



Console screenshot — A8: “What is the quality target for the usage-to-subscription match rate, and what triggers an investigation?”

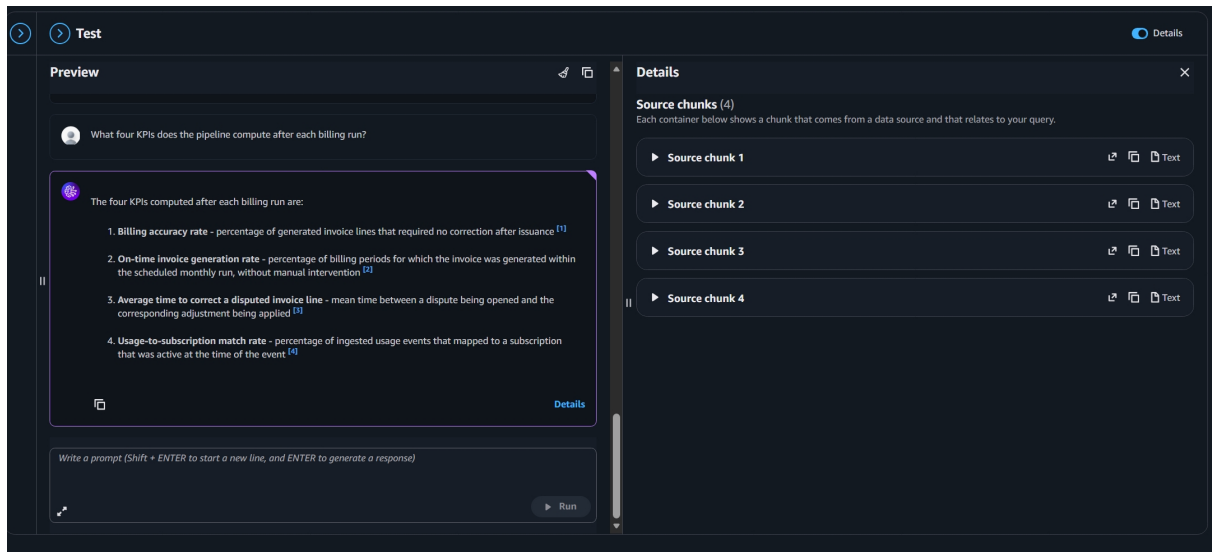
grounding_result: UNGROUNDED_RISK

failure_analysis: The underlying facts retrieved are correct (99.5% target; investigation triggered when the match rate is sustained below 99% for 3 days), and two citations are present. But the generated text itself is garbled and duplicated — "99.5" is split into "99." and a stray "5", and the investigation-trigger sentence is repeated twice, cut short mid-number both times. As displayed, the answer is not reliably readable even though retrieval found the right chunks — a generation defect, not a retrieval defect.

A9 · Answerable

Question: What four KPIs does the pipeline compute after each billing run?

Expected source: pipeline_README.md



Console screenshot — A9: “What four KPIs does the pipeline compute after each billing run?”

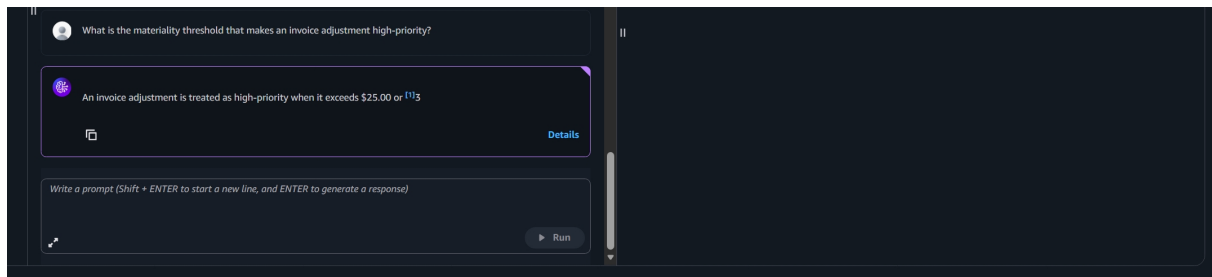
grounding_result: GROUNDED

failure_analysis: Lists billing accuracy rate, on-time invoice generation rate, average time to correct a disputed invoice line, and usage-to-subscription match rate, each with its own citation (Source chunks (4) panel confirms four distinct retrieved chunks).

A10 · Answerable

Question: What is the materiality threshold that makes an invoice adjustment high-priority?

Expected source: data_quality_rules.md



Console screenshot — A10: “What is the materiality threshold that makes an invoice adjustment high-priority?”

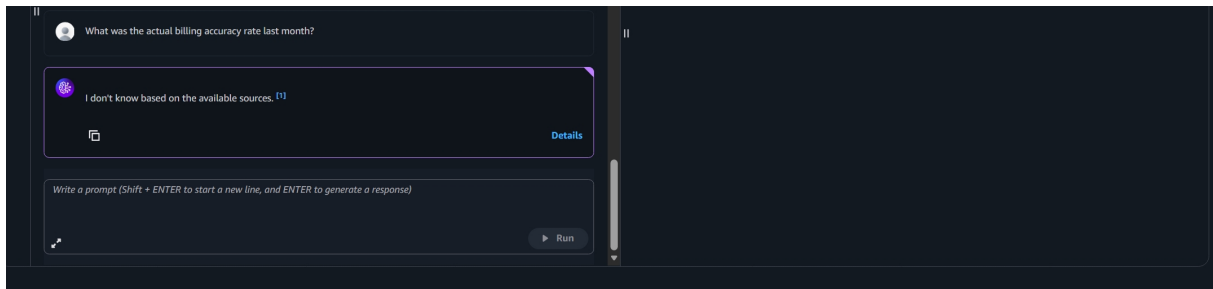
grounding_result: UNGROUNDED_RISK

failure_analysis: The answer is cut off mid-sentence: "...exceeds \$25.00 or [1]3" — the "5% of the original invoice amount, whichever is greater" clause is missing entirely, leaving only a stray floating "3" where the number should continue. This is a materially incomplete answer: a reader would not learn about the 5%-of-invoice alternative criterion at all. Same class of generation truncation/garbling seen in A8.

U1 · Unanswerable

Question: What was the actual billing accuracy rate last month?

Expected source: none (runtime data, not in the document set)



Console screenshot — U1: "What was the actual billing accuracy rate last month?"

grounding_result: ABSTAINED

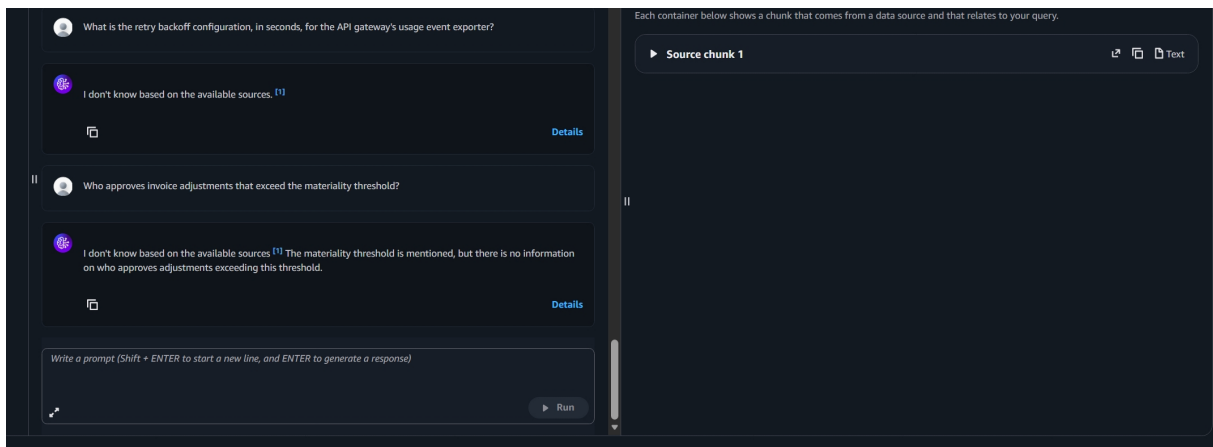
failure_analysis: Correctly abstained with "I don't know based on the available sources." — no runtime figure was fabricated. Minor artifact: a [1] citation marker is attached to the abstention even though nothing was actually cited to produce it.

U2 · Unanswerable

Question: What is the retry backoff configuration, in seconds, for the API gateway's usage event exporter?

Expected source: none (a different system's implementation detail, never specified)

Both remaining unanswerable questions were run in the same session; one screenshot captures both exchanges.



Console screenshot — U2: "What is the retry backoff configuration, in seconds, for the API gateway's usage event exporter?"

grounding_result: ABSTAINED

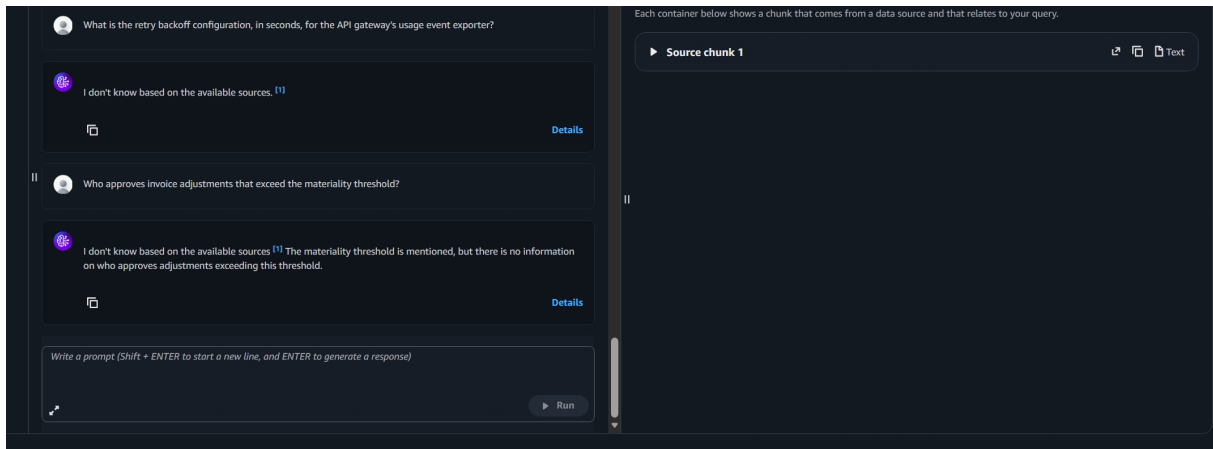
failure_analysis: Correctly abstained rather than inventing a plausible-sounding number of seconds. Same minor citation-on-abstention artifact as U1.

U3 · Unanswerable

Question: Who approves invoice adjustments that exceed the materiality threshold?

Expected source: none (personnel/approval-workflow information not in the document set)

Same screenshot as U2, second exchange.



Console screenshot — U3: “Who approves invoice adjustments that exceed the materiality threshold?”

grounding_result: ABSTAINED

failure_analysis: Correctly abstained, and — usefully — added that the materiality threshold itself is mentioned in the source material but no approver/role is specified, rather than guessing a name or title. This is the strongest abstention response of the three: it distinguishes what is and isn't covered instead of a bare refusal.

4. Overall analysis

Grounding, across all 13 questions: 10/10 answerable questions (A1–A10) returned content that is factually grounded in the correct source document, each with at least one citation into that document. 3/3 unanswerable questions (U1–U3) correctly abstained rather than guessing or fabricating an answer — no hallucination occurred anywhere in this run. By the strict definitions in Section 3, that is a 10/10 GROUNDED result on answerable questions and a 3/3 ABSTAINED result on unanswerable questions, which is the core pass/fail signal this assessment is designed to measure.

Residual defects, all on the generation side rather than retrieval: three responses (A1 pre-fix, A8, A10) show garbled, duplicated, or truncated text around numeric values, even though the correct source chunks were retrieved and cited in every case. A10 in particular drops a substantive clause (the 5%-of-invoice-amount alternative), which would mislead a reader relying only on the displayed answer. This pattern is concentrated in Nova Micro's handling of longer, number-heavy sentences and did not recur on short, non-numeric answers (A2, A6, A7, A9). Separately, two abstentions (U1, U2) carry a stray [1] citation marker despite citing nothing, a minor template/model quirk rather than a grounding failure.

Recommendation: retrieval, chunking, and the prompt template's grounding/abstention rules are working as designed — the vector store is returning the right chunks for every question, including the three designed-to-fail unanswerable ones. The open item is generation quality on Nova Micro for numeric, multi-clause answers (A8, A10). Re-running just those two questions against a stronger generation model (e.g. Claude 3.5 Haiku) while keeping the same Knowledge Base and prompt template would confirm whether this is a Nova Micro-specific limitation before treating the pipeline as production-ready.